

21. In the modern periodic table, one of the following does not have an appropriate position:

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- (a) transition elements
- (b) inert gases
- (c) inner transition elements
- (d) halogens

Answer: (c) inner transition elements

Explanation: The f-block (inner transition) elements are shown separately at the bottom of the usual modern periodic table to keep the table compact. Hence their displayed position is not fully integrated into the main body.

22. Element E belongs to the 4th period and group 16 of the periodic table. The valence-shell electron configuration of the element just above E in the group is

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- (a) $3s^2 3p^4$
- (b) $3s^2 3p^5$
- (c) $3s^2 3p^6$
- (d) $3s^2 3p^3$

Answer: (a) $3s^2 3p^4$

Explanation: An element in period 4, group 16 is Se. The element immediately above Se is S, whose valence-shell configuration is $3s^2 3p^4$.

23. The electronic configurations of four elements are given below. Which element does not belong to the same family as the others?

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- (a) $[\text{Xe}] 4f^{14} 5d^{10} 6s^2$
- (b) $[\text{Kr}] 4d^{10} 5s^2$
- (c) $[\text{Ne}] 3s^2 3p^5$
- (d) $[\text{Ar}] 3d^{10} 4s^2$



Answer: (c) $[\text{Ne}] 3s^2 3p^5$

Explanation: Options (a), (b) and (d) correspond to Hg, Cd and Zn, all group 12 elements. $[\text{Ne}]3s^23p^5$ is chlorine, a group 17 halogen, so it belongs to a different family.

24. The elements with atomic numbers 9, 17, 35, 53 and 85 belong to

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- (a) alkali metals
- (b) alkaline earth metals
- (c) halogens
- (d) noble gases

Answer: (c) halogens

Explanation: Atomic numbers 9, 17, 35, 53 and 85 correspond to F, Cl, Br, I and At. These are group 17 elements, called halogens.

25. Which of the following pairs has both members from the same period of the periodic table?

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- (a) Na – Ca
- (b) Na – Cl
- (c) Ca – Cl
- (d) Cl – Br

Answer: (b) Na – Cl

Explanation: Na ($Z = 11$) and Cl ($Z = 17$) are both in period 3. The other pairs contain elements from different periods.

26. The elements characterized by the outer electronic configuration ns^1 to ns^2np^6 are collectively called

- (a) transition elements
- (b) representative elements
- (c) lanthanides
- (d) inner transition elements



Answer: (b) representative elements

Explanation: Elements whose outer configurations range from ns^1 through ns^2np^6 are the s- and p-block elements, collectively called representative or main-group elements.

27. f-Block elements are called inner transition elements because

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- (a) they have properties similar to those of transition elements
- (b) they exist in between transition elements
- (c) the last electron enters into the f-orbital of the penultimate shell
- (d) the last electron enters into any orbital of the penultimate shell

Answer: (c) the last electron enters into the f-orbital of the penultimate shell

Explanation: In f-block elements the differentiating electron enters an inner f-subshell, commonly represented as $(n-2)f$. This inner filling is the reason they are called inner transition elements; among the given choices, (c) expresses the intended idea.

28. An element which belongs to the third period and group 16 of the periodic table has electronic configuration

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- (a) $1s^2 2s^2 2p^6 3s^2 3p^3$
- (b) $1s^2 2s^2 2p^6 3s^2 3p^4$
- (c) $1s^2 2s^2 2p^6 3s^2 3p^5$
- (d) $1s^2 2s^2 2p^4$

Answer: (b) $1s^2 2s^2 2p^6 3s^2 3p^4$

Explanation: Period 3 means the highest principal shell is $n = 3$. Group 16 p-block elements have valence configuration ns^2np^4 ; therefore the full configuration is $1s^2 2s^2 2p^6 3s^2 3p^4$.

29. * In the long form of the periodic table, all the non-metals are placed in the

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- (a) s-block
- (b) p-block



- (c) f-block
- (d) d-block

Answer: (b) p-block

Explanation: The great majority of non-metals lie on the right side of the periodic table in the p-block. This is the intended NCERT-based answer (with hydrogen being the familiar s-block exception).

30. * Which period contains most of the man-made radioactive elements?

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- (a) Seventh
- (b) Fifth
- (c) Sixth
- (d) Both (a) and (c)

Answer: (a) Seventh

Explanation: Most synthetic, highly radioactive transuranium elements are in period 7, especially in and beyond the actinoid series.

31. The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^3$. What is the atomic number of the element just below the above element in the periodic table?

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- (a) 33
- (b) 34
- (c) 36
- (d) 49

Answer: (a) 33

Explanation: The given configuration is phosphorus, $Z = 15$, a group 15 element. The element directly below phosphorus is arsenic, whose atomic number is 33.

32. * An element X belongs to the fourth period and fifteenth group of the periodic table. Which one of the following is true regarding the outer electronic configuration of X? It has

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- (a) partially filled d-orbitals and completely filled s-orbitals
- (b) completely filled s-orbital and completely filled p-orbitals
- (c) completely filled s-orbital and half-filled p-orbitals
- (d) half-filled d-orbitals and completely filled s-orbitals

Answer: (c) completely filled s-orbital and half-filled p-orbitals

Explanation: Period 4, group 15 corresponds to As with valence configuration $4s^2 4p^3$. The s orbital is filled and the three p orbitals are half-filled.

33. * An element has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^4$. Its period, block and group are

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- (a) Period 3, p-block, group 16
- (b) Period 5, s-block, group 1
- (c) Period 3, p-block, group 10
- (d) Period 4, d-block, group 12

Answer: (a) Period 3, p-block, group 16

Explanation: The highest principal quantum number is 3, so the element is in period 3. The last electron enters a p-orbital, making it p-block, and $ns^2 np^4$ corresponds to group 16.

34. An element having atomic number 56 belongs to which block of the periodic table?

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- (a) p-block
- (b) d-block
- (c) f-block
- (d) s-block

Answer: (d) s-block

Explanation: Atomic number 56 is barium, Ba, with outer configuration $6s^2$. Therefore it belongs to the s-block.



35. Which of the given elements A, B, C, D and E, with atomic numbers 2, 3, 7, 10 and 30 respectively, belong to the same period?

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- (a) A, B, C
- (b) B, C, D
- (c) A, D, E
- (d) B, D, E

Answer: (b) B, C, D

Explanation: $Z = 3$ (Li), 7 (N) and 10 (Ne) all have electrons occupying the second shell as their highest occupied shell, so all are in period 2.

36. Which of the following elements forms a basic oxide?

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- (a) Aluminium
- (b) Carbon
- (c) Nitrogen
- (d) Magnesium

Answer: (d) Magnesium

Explanation: Magnesium forms MgO , a predominantly ionic and basic oxide. Al_2O_3 is amphoteric, whereas oxides of carbon and nitrogen are generally acidic or neutral depending on oxidation state.

37. Elements of which group form anions most readily?

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- (a) Oxygen family
- (b) Nitrogen family
- (c) Halogens
- (d) Alkali metals



Answer: (c) Halogens

Explanation: Halogens have seven valence electrons and high electron-gain tendency. They readily gain one electron to form X^- anions.

38. Which of the following is non-metallic?

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- (a) B
- (b) Be
- (c) Mg
- (d) Al

Answer: (a) B

Explanation: Among the given choices, boron has the greatest non-metallic character and is classified as a metalloid. Be, Mg and Al are metals.

39. The only non-metal which is liquid at ordinary temperature is

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- (a) Hg
- (b) Br
- (c) Na
- (d) None of these

Answer: (b) Br

Explanation: Bromine is the only non-metal that is liquid at ordinary room temperature. Mercury is also liquid, but it is a metal.

40. Arrange the following elements in the order of increasing non-metallic character: Li, O, C, Be and F.

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- (a) $F < O < C < Be < Li$
- (b) $Li < Be < C < O < F$
- (c) $F < O < C < Be < Li$
- (d) $F < O < Be < C < Li$



Answer: (b) $\text{Li} < \text{Be} < \text{C} < \text{O} < \text{F}$

Explanation: Across period 2, effective nuclear charge increases and metallic character decreases, so non-metallic character increases from left to right. Hence $\text{Li} < \text{Be} < \text{C} < \text{O} < \text{F}$.

